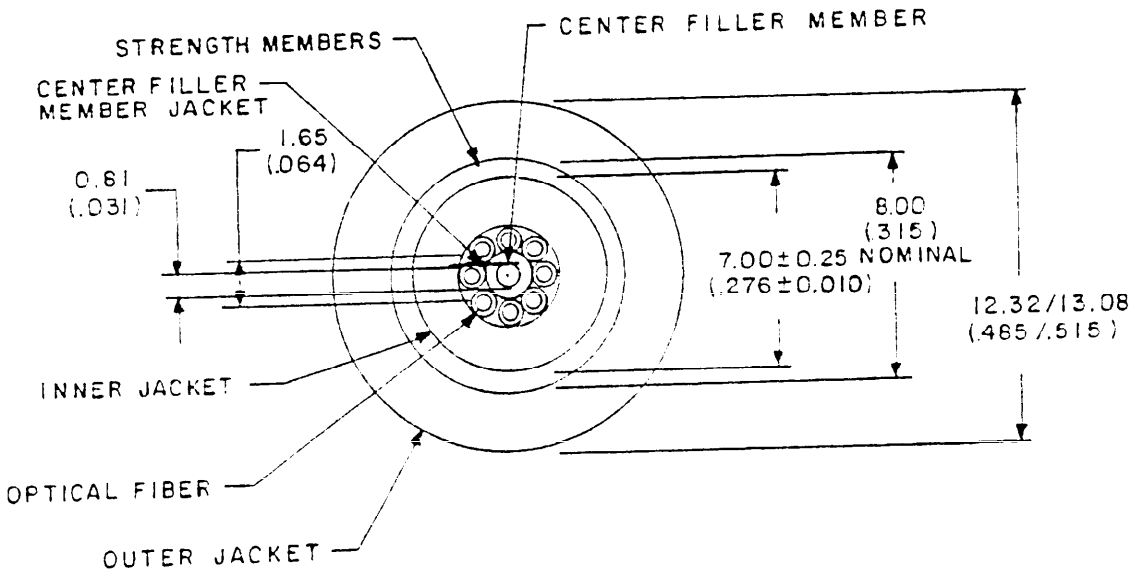
MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTICS, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of DOD-C-85045.



NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Inches are in parentheses.

FIGURE 1. Eight fiber cable.

AMSC N/A FSC 6015

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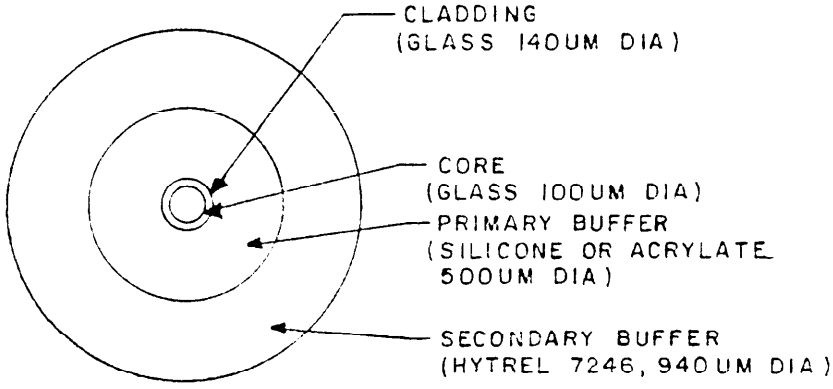


FIGURE 2. Optical fiber detail.

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REQUIREMENTS:

Cable construction detail: See figure 1.

Cable optical fiber lay: 7.6 cm.

Cable outer strength member lay: Helically served.

Cable jacket color: Black.

Fiber color code: The Hytrel buffer shall code the fibers with the following colors - brown, red, orange, grey, green, blue, black, and white.

Optical fiber detail: Optical fiber parameters shall be measured prior to cabling (see figure 2).

Fiber core diameter: 100 ±3 micrometers measured across any diameter using test method EIA-455-43.

Fiber cladding diameter: 140 ±2 micrometers measured across any diameter using test method EIA-455-27.

Core-cladding offset: < 5 micrometers using EIA-455-45.

Coating diameter: 500 ±25 micrometers using test method EIA-455-55.

Coating to cladding offset: < 25 micrometers using test method EIA-455-55.

Buffer diameter: 940 ±50 micrometers.

Coated fiber-buffer offset: < 25 micrometers.

Tensile strength proof level: .6895 GPa using test method EIA-455-31.

Materials:

Fiber:

Core - Glass.

Cladding - Glass.

Coating - Silicone, epoxy acrylate, or equivalent.

Buffer: Hytrel 7246, or equivalent.

Center filler member: Air Logistics Corporation part number 118633, or equivalent.

Outer jacket concentricity: 0.8 minimum.

Cable mass per unit length: 220 kg/km.

Cable optical performance:

Attenuation: $\leq$ 6.0 dB/km at 850 ±30 nm wavelength.

Bandwidth: $\geq$ 20 MHz-km at 850 nm ±20 nm.

Numerical aperture: 0.290 ±0.015 defined at the 5 percent intensity level for a 2.0 ±0.1 meter fiber sample length. Fiber source end illumination shall be uniform over the core with a numerical aperture of 0.35 ±0.01; without index matching fluid. All fibers in a cable shall be tested. Acceptance pattern measurement is not required. Use test method EIA-455-31.

Index profile: g > 20 with no center dip compensation using test method EIA-455-44. Variation in index across the core diameter from 0.1 radius to 0.85 radius shall be less than ±5 percent.

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Jacket opacity: Shall be verified at 850 nm ±50 nm.

Environmental performance:

Temperature cycling (test to be done with cable on reel): After being subjected to 5 cycles of temperature cycling, the induced attenuation shall not exceed 0.5 dB. Temperature are -62°C ±2°C and +50°C ±2°C. Soak times at temperature extremes and at +25°C ±2°C shall be four hours minimum.

Low temperature flexibility (cold bend): The test temperature shall be -20°C and the induced attenuation shall not exceed 1 dB.

Cyclic flexing: The mandrel diameter shall be ten times the cable outer diameter.

Tensile loading and elongation: The maximum tensile test load shall be 1,200 newtons. Maximum allowable elongation shall not exceed 1 percent.

Fluid immersion: The fluids shall be limited to lubricating oil, hydraulic oil, and salt solution. After testing the cable diameter shall not exceed the maximum cable diameter specified.

Flammability: Vertical burn procedure.

Corner bend: The force shall be 75 newtons.

Operating temperature range: -20°C to +50°C.

Storage temperature range: -62°C to +71°C.

Cold working: The chamber temperature shall be 0°C ±2°C.

Engineering information:

Cable minimum bend radius: 127 mm (5 inches).

Part number : D85045/06-C8B.

Parameter applicability: See table I.

TABLE II. Parameter applicability. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Group I					
Visual and mechanical	6 units 1 km each	X	X		
Attenuation rate	6 units 1 km each	X	X		
Optical fiber bandwidth	6 units 1 km each	X	X		
Crosstalk	6 units 1 km each	X	X		
Numerical aperture	6 units 2 m each (1 each from 6 cables)	X	X		
Group II					
Temperature cycling	6 units 1 km each (3 on reel, 3 off reel)	X		X	
Thermal shock	2 units 1 km each (1 on reel, 1 off reel)				

See footnotes at end of table.

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TABLE I. Parameter applicability - Continued. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Humidity	2 units 1 km each	X		X	
Storage temperature	1 units 1 km each (1 on reel, 1 off reel)	X		X	
Barometric pressure (reduced)	1 unit 1 km				
Life at elevated ambient temperature	2 units 500 m each				
Tensile loading	6 units 150 m each (2 each from 3 cables)				
Operational tensile loading	2 units 150 m each (1 each from 2 cables)				
Freezing water immersion (ice crush)	2 units 30 m each (1 each from 2 cables)	X		X	
Fungus	2 units 2 m each (1 each from 2 cables)	X		X 3/	
Group III (Military environments)					
Low temperature flexibility (cold bend)	3 units 8 m each (1 each from 3 cables)	X			X
Cyclic flexing	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Crush	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Cable twist-bending	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Impact	9 units 5 m each (3 each from 3 cables)	X			X
Jacket self-adhesion or blocking	2 units ≥5 m each (1 each from 2 cables)	X			X
Wicking	2 units 2 m each (1 each from 2 cables)	X			X
Cable jacket opacity	3 units 2 m each (1 each from 2 cables)	X			X

See footnotes at end of table.

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Insp	Quality assurance group A	group B	group C
Fluid immer			X
Corner bend			X
Knot			X
Ultraviolet			X
Cable eleme			X
Durability identific			X
Flammabilit			X
Group I			
Hosing - low pre			X
- hydrost			X
Dripping			X
Oxygen pres			X
Gas flame			X
Pressure cy			X
Cold workin			X
Stuffing tu			X
Acid gas ge			X
Halogen con			X
Limiting ox			N/A
Smoke index			X
Toxicity in			X

1/ Toleran normalized to 1 km.

2/ Toleran

3/ Fungus purposes.

C

±50 nm.

ole on reel): After being subjected
ged attenuation shall not exceed 0.5
°C. Soak times at temperature
rs minimum.

e test temperature shall be -20°C and
3.

be ten times the cable outer diameter.

tensile test load shall be 1,200
not exceed 1 percent.

d to lubricating oil, hydraulic oil,
diameter shall not exceed the maximum

be 0°C ±2°C.

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ability. 1/ 2/

Qualification	Quality assurance group A	group B	group C
X	X		
X	X		
X	X		
X	X		
X	X		
X		X	
)			
)			

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Navy - SH

Agent:
DLA - ES

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